

LINMA2222 Project Presentation

Optimal Portfolio Strategy

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Outline

1. Problem & model	3	3.1 Optimizing linear and	
1.1 Goal	4	quadratic policies	13
1.2 Dynamics (true system) . .	5	4. Model-based control (LQR /	
1.3 Reward (interpretation) ...	6	MPC)	15
2. Baselines & constraints	7	4.1 LQR approximation	16
2.1 Baseline linear policy		4.2 LQR optimal policy	
(unclipped)	8	(unclipped)	17
2.2 Constrained baseline		4.3 Constraints: clipped LQR vs	
(clipping)	10	MPC	19
3. Direct policy search (CMA-		5. RL / ADP methods	21
ES)	12	5.1 LSTD $\rightarrow$ LSPI	22
		5.2 Q- λ learning	23

5.3 Poisson error & LSPE(+PI)	24
6. Conclusion	26
6.1 Summary	27
6.2 Design choices & difficulties	28
6.3 Final slide	29
7. Appendix	30
7.1 Extra: reward quadratic form (if asked)	31
7.2 Extra: baseline distributions	32

1. Problem & model

Goal

- Control trades u_t to manage inventory q_t and maximize long-run performance.
- Compare:
 - baseline linear policies (unconstrained vs constrained),
 - direct policy search (CMA-ES),
 - model-based control on an LQR approximation (LQR / clipped LQR / MPC),
 - RL methods (LSTD/LSPI, Q - λ , Poisson error + LSPE/PI).

Realism constraint: $q_t \in [0, 1]$ (no shorting, bounded inventory).

Dynamics

State $x_t = (q_t, z_t^a, z_t^u)^\top$

Control u_t

Noise $\xi_t^a, \xi_t^p \sim \mathcal{N}(0, 1)$

$$q_{t+1} = q_t + u_t$$

$$z_{t+1}^a = (1 - \omega^a) z_t^a + \omega^a \sigma^a \xi_t^a$$

$$z_{t+1}^u = (1 - \omega^u) z_t^u + \omega^u \beta^u u_t$$

Price update used inside reward: $p_{t+1} = p_t + z_t^a + z_t^u + \gamma^u u_t + \sigma^p \xi_t^p$

1.3 Reward (interpretation)

Reward = step P/L

$$g_t := 1000 \cdot \left((q_{t+1}p_{t+1} - q_t p_t) - (q_{t+1} - q_t) \bar{p}_{t,t+1} \right)$$

- First term: mark-to-market change of holdings.
- Second term: cash flow at execution price $\bar{p}_{t,t+1}$.
- So g_t is the (gross) profit/loss from t to $t + 1$.

Utilities:

- $c_{\text{quad}}(g) = g - \frac{1}{2}g^2$
- $c_{\text{exp}}(g) = 1 - \exp(-g)$

2. Baselines & constraints

2.1 Baseline linear policy (unclipped)

Baseline policy π_{cl}

$$\pi_{\text{cl}}(x_t) = K_{\text{cl}}x_t = -0.5q_t + 0.5z_t^a + 0.5z_t^u$$

What we observe:

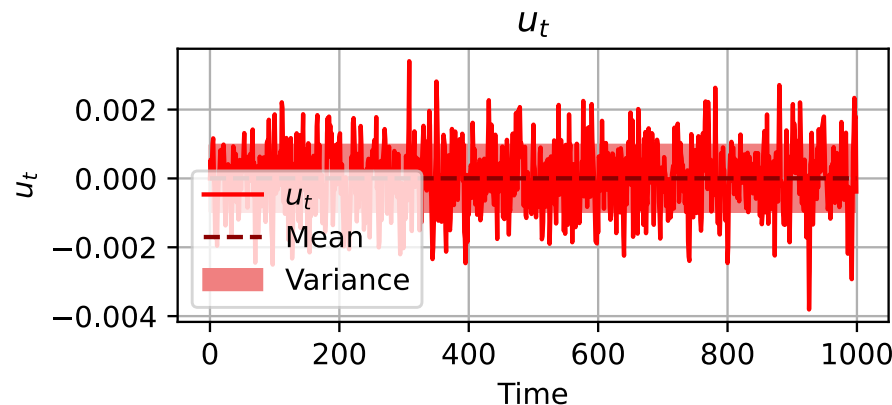
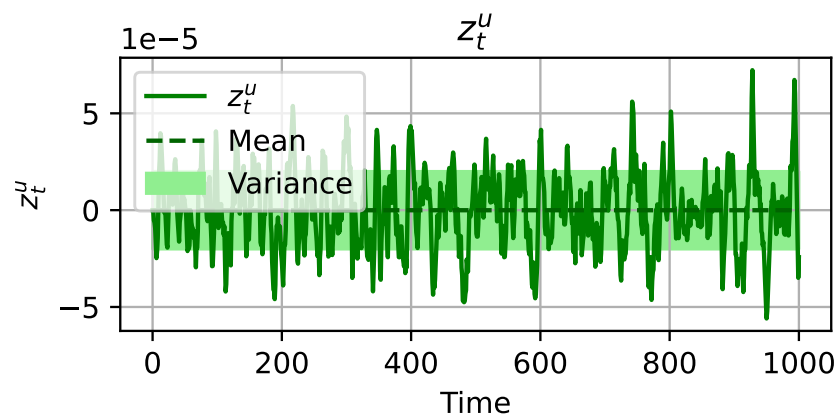
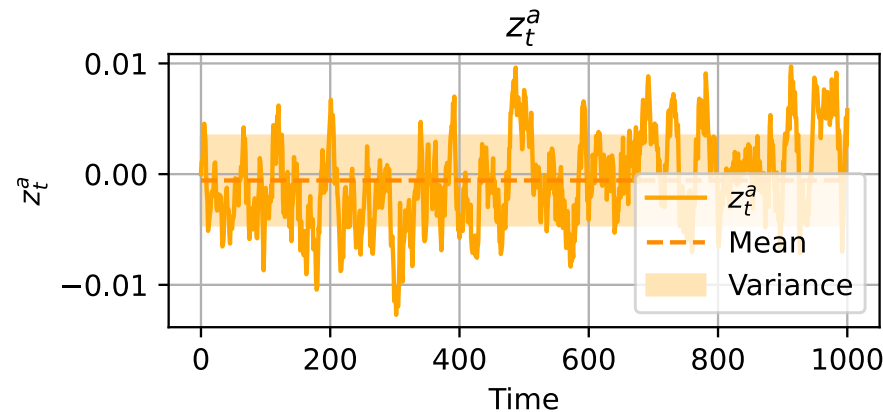
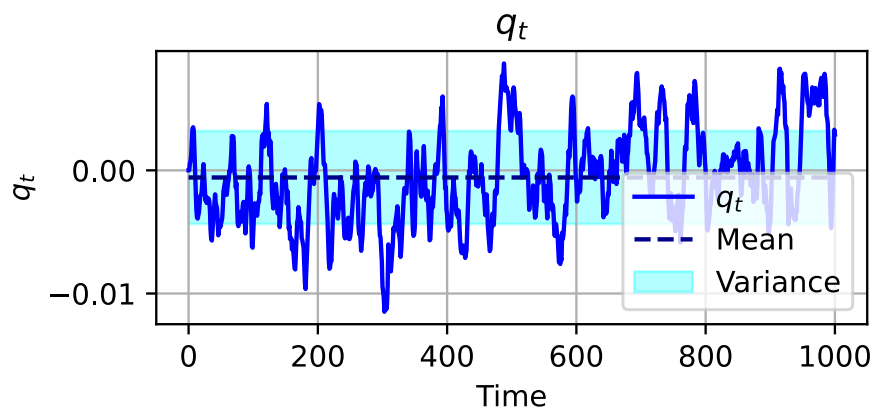
- states/actions centered around 0,
- but q_t can become negative (shorting),
- long-run average reward $> 0 \rightarrow$ profitable but unrealistic.

2.1 Baseline linear policy (unclipped)

2. Baselines & constraints

Trajectories of states and actions over time

Policy: unclipped policy



Clipped baseline policy π_{pcl}

Enforce $q_t \in [0, 1]$ by clipping actions: $\pi_{\text{pcl}}(x_t) = \text{clip}(\pi_{\text{cl}}(x_t), [-q_t, 1 - q_t])$

Effects:

- Feasible inventory always.
- “Plateaux” when controller saturates (wants illegal trades).
- Average reward drops during saturation episodes.

2.2 Constrained baseline (clipping)

2. Baselines & constraints

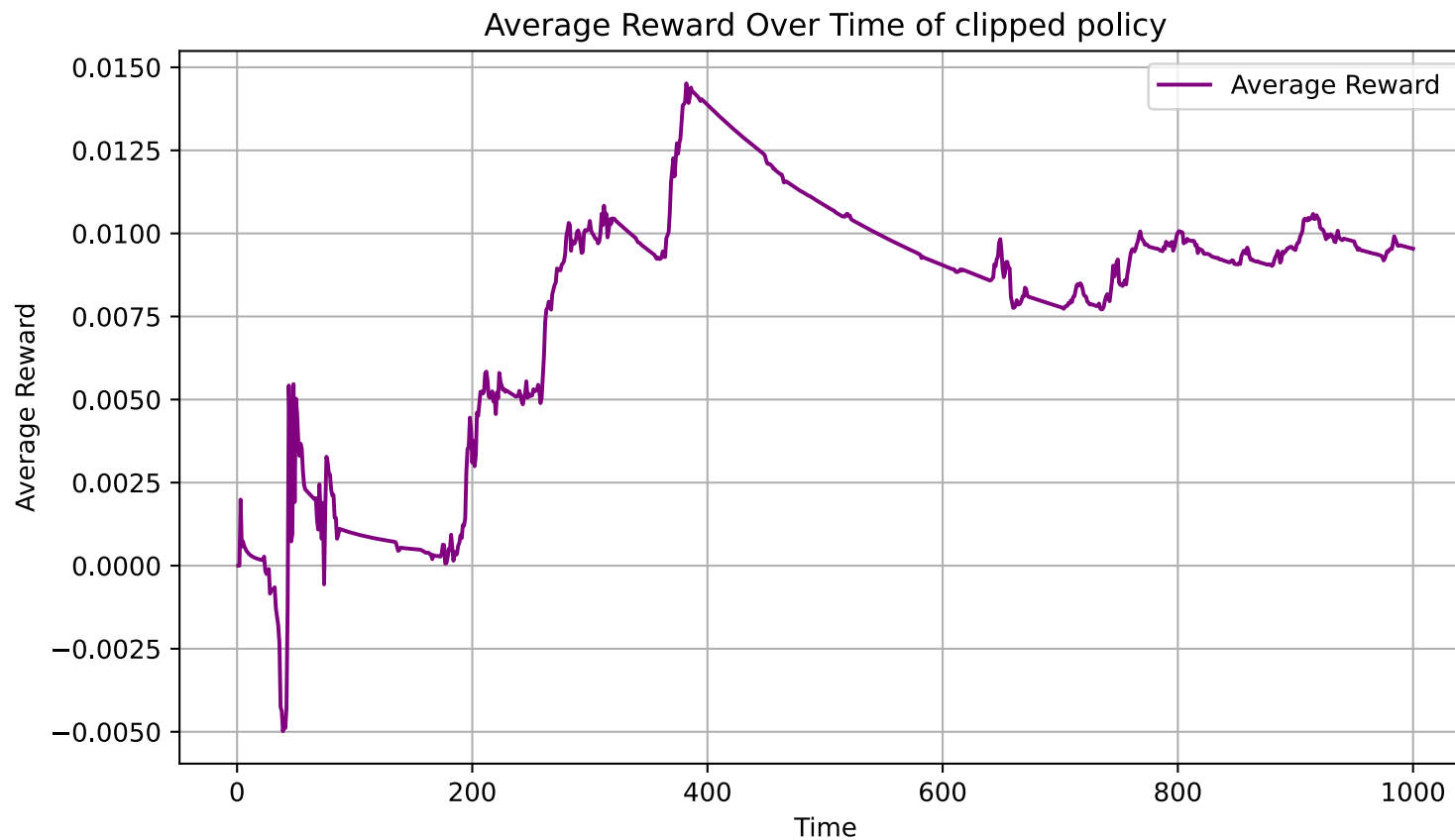


Figure 2: Average reward drops when clipping is active

3. Direct policy search (CMA-ES)

3.1 Optimizing linear and

Quadratic policies CMA-ES policy search

Linear clipped form: $\pi_{l(x)} = \text{clip}\left(p_3[q_t, z_t^a, z_t^u]^\top, [-q_t, 1 - q_t]\right)$

Quadratic clipped form: $\pi_{q(x)} = \text{clip}(p_{10}\varphi(x_t), [-q_t, 1 - q_t])$

Empirical takeaway:

- CMA-ES improves reward distributions vs baselines.
- Quadratic slightly better than linear $\rightarrow$ diminishing returns.

3.1 Optimizing linear and quadratic policies

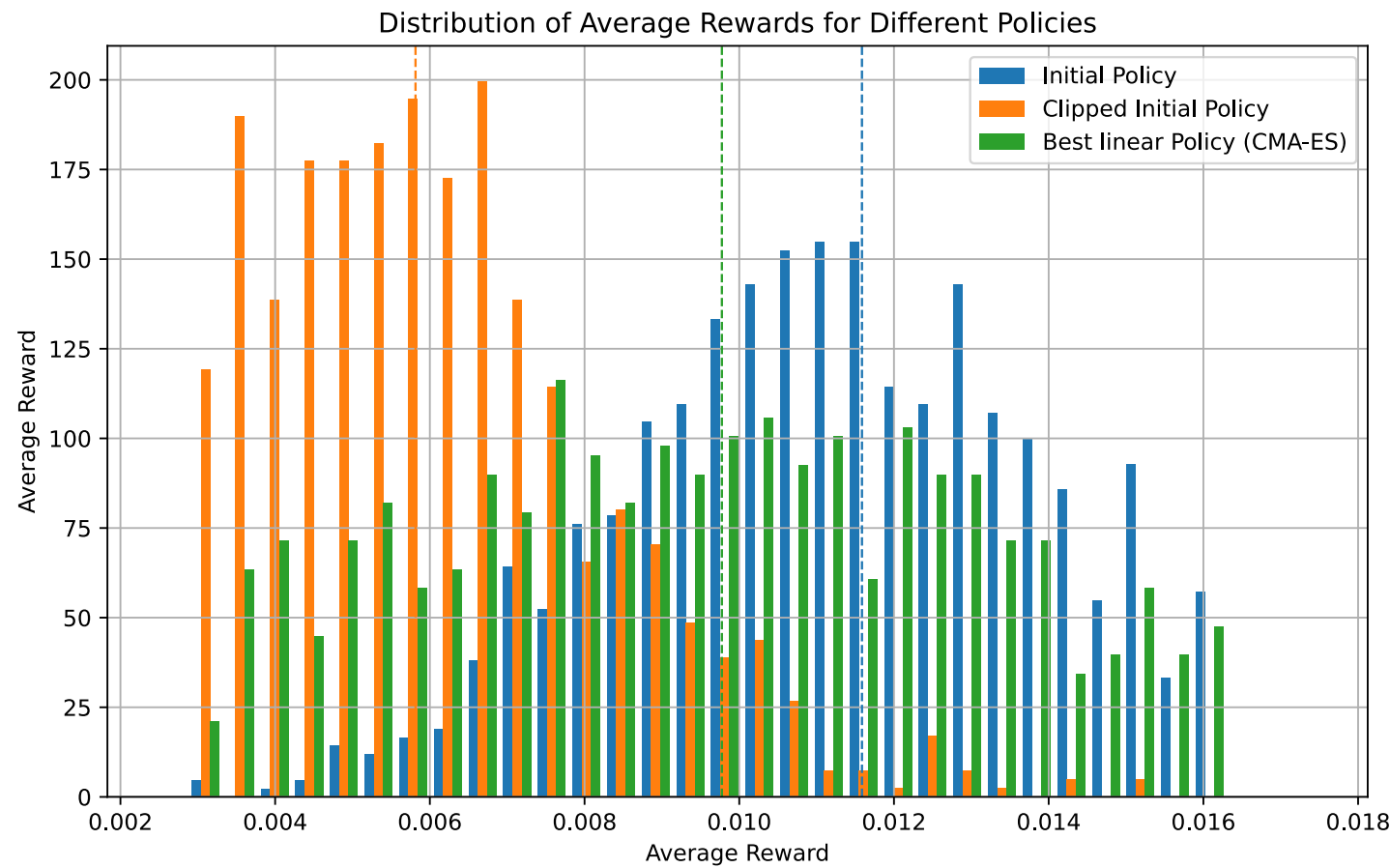


Figure 3: Reward comparison: baseline vs optimized

4. Model-based control (LQR / MPC)

LQR approximation

Linear dynamics: $x_{t+1} = Fx_t + Gu_t + D\xi_t$

$$\text{with } F = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1-\omega^a & 0 \\ 0 & 0 & 1-\omega^u \end{pmatrix} \text{ and } G = \begin{pmatrix} 1 \\ 0 \\ \omega^u \beta^u \end{pmatrix}.$$

Reward approximation (2nd order): $\mathbb{E}[c_{\text{quad}}(g_t) \mid x_t, u_t] \approx \frac{1}{2}x^\top Sx + x^\top Pu + \frac{1}{2}u^\top Ru$

Convert to cost minimization (standard infinite-horizon LQR).

4.2 LQR optimal policy

(unclipped)
LQR policy π_{lqr}

Riccati solution gives: $K_{\text{lqr}} \approx (1.112 \quad -2.649 \quad -2.528)$

$$\pi_{\text{lqr}}(x_t) = -K_{\text{lqr}}x_t$$

Observations:

- More aggressive actions and larger inventory swings.
- Much higher average reward, close to theoretical prediction.

4.2 LQR optimal policy

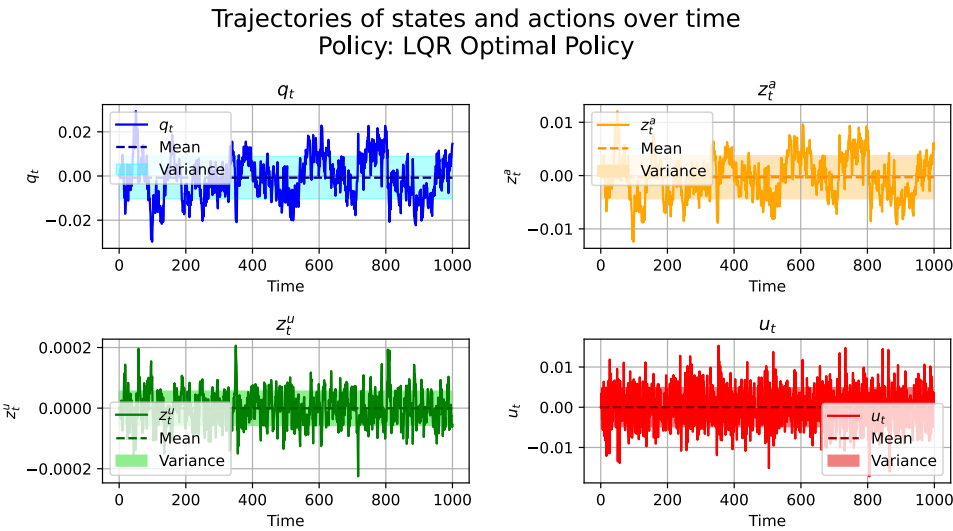


Figure 4: States & action

4. Model-based control (LQR / MPC)

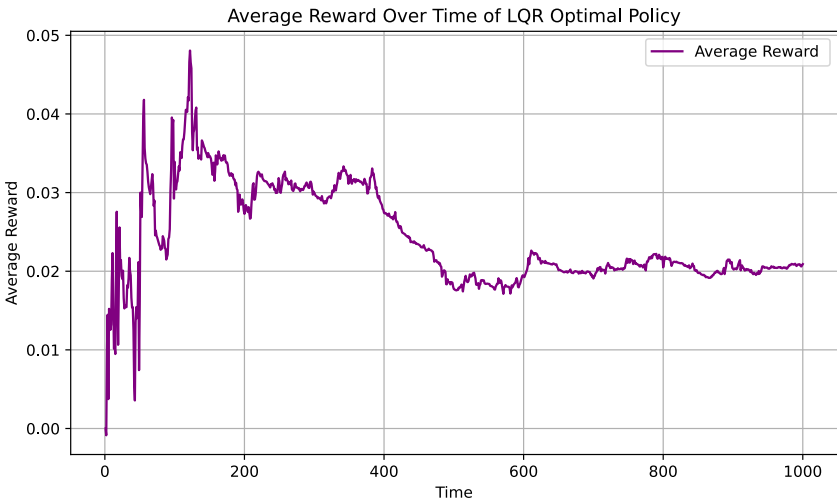


Figure 5: Average reward

Clipped LQR and MPC

- Clipped LQR (feasible but saturates) $\rightarrow$ reward decreases.
- MPC solves a constrained optimization each step:
 - choose horizon $\mathcal{N}$ (we used $\mathcal{N} = 10$),
 - similar performance to clipped LQR in our experiments,
 - higher computation cost.

4.3 Constraints: clipped LQR vs 4. Model-based control (LQR / MPC)

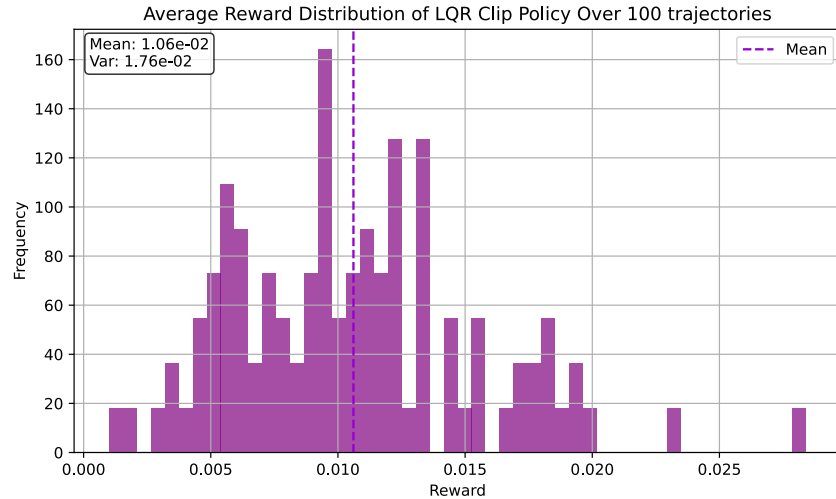


Figure 6: Clipped LQR reward distribution

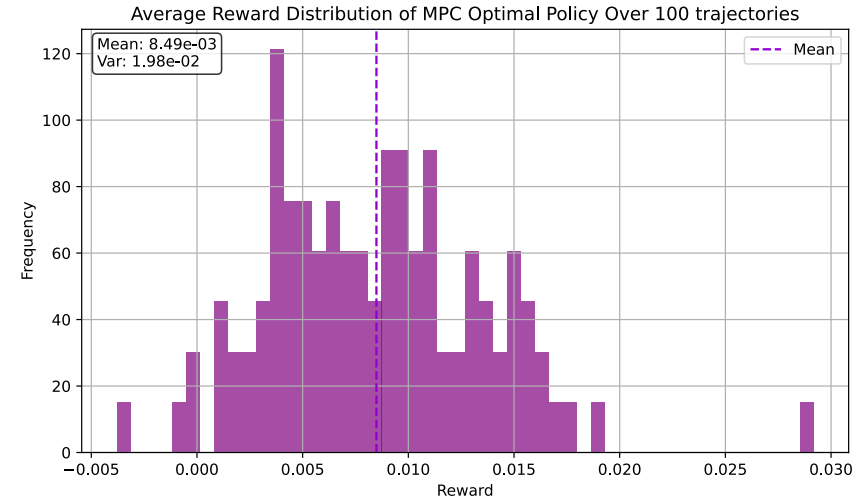


Figure 7: MPC reward distribution

5. RL / ADP methods

5.1 LSTD $\rightarrow$ LSPI

LSTD and LSPI

- Approximate Q by $Q^{\theta(x,u)} \approx \theta^\top \psi(x, u)$.
- Basis choice:
 - quadratic basis matches LQR structure exactly,
 - higher-degree basis needed for non-quadratic utilities.

Result:

- LSPI improves over π_{cl} , and can approach model-based performance.

5.2 Q- λ learning

Q- λ learning

- Learned gains move toward K_{lqr} (on approximate system).
- Larger λ speeds learning but may become unstable unless step size is reduced.

5.3 Poisson error & LSPE(+PI)

Poisson error, LSPE, and policy improvement

Average-reward setting uses Poisson error: $\mathcal{P}^{\theta(x,u)} = \mathbb{E} \left[r + Q^{\theta(x^+, \pi(x^+))} \mid x, u \right] - \eta^\theta - Q^{\theta(x,u)}$

- Minimizing MSPE $\rightarrow$ least squares in (θ, η) .
- With a finite-support kernel (Monte Carlo) we can approximate the expectation.

With policy improvement (LSPE+PI):

- start from π_{cl} ,
- alternates evaluation and improvement,

5.3 Poisson error & LSPE(+PI)

- can recover LQR-like behavior.

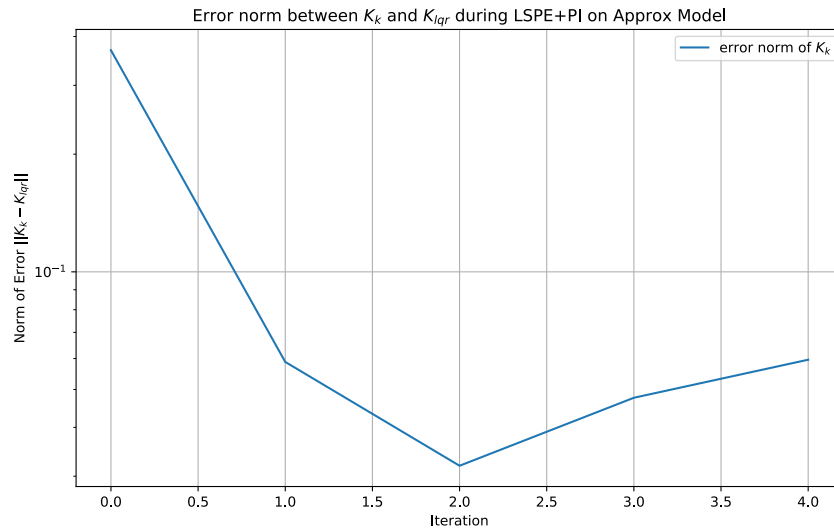


Figure 8: Convergence to K_{lqr}

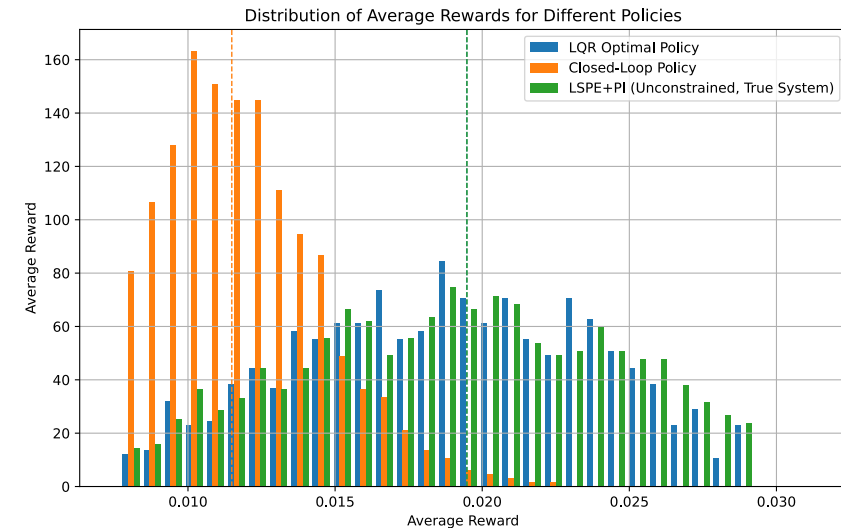


Figure 9: Rewards: π_{cl} vs π_{lqr} vs π_{lspepi}

6. Conclusion

6.1 Summary

Key takeaways

- Best unconstrained controller: π_{lqr} (highest average reward, but allows $q_t < 0$).
- Best feasible controllers:
 - clipped LQR $\approx$ MPC (in our experiments),
 - MPC costs more compute without clear gains.
- RL/ADP:
 - with the right basis, LSPI and LSPE+PI recover near-LQR performance quickly,
 - constraint handling is the main difficulty and source of performance loss.

6.2 Design choices & difficulties

Design choices and difficulties

Design choices

- action clipping to enforce $q_t \in [0, 1]$,
- CMA-ES for noisy objective optimization,
- quadratic basis for LQR-consistent learning,
- MPC horizon $\mathcal{N} = 10$ as compute/reward compromise.

Difficulties

- clipping saturation reduces reward during “good opportunities”,
- Q- λ instability at high λ without tuning α ,
- average-reward evaluation requires estimating $\eta \rightarrow$ Poisson/LSPE is natural.

Conclusion

- LQR gives a strong, interpretable benchmark and near-theoretical performance.
- With constraints, clipped LQR and MPC provide feasible control (similar performance here).
- RL methods can match model-based control when approximation architecture matches structure.

Backup slides in appendix.

7. Appendix

7.1 Extra: reward quadratic form (if asked)

Reward as a quadratic form

You derived: $g_t = \frac{1}{2} y_t^\top H y_t$ with $y_t = (q_t, z_t^a, z_t^u, u_t, \xi_t^a, \xi_t^p)^\top$.

Use this slide when asked “why quadratic approximations make sense here?”.

7.2 Extra: baseline distributions

Extra: reward distributions

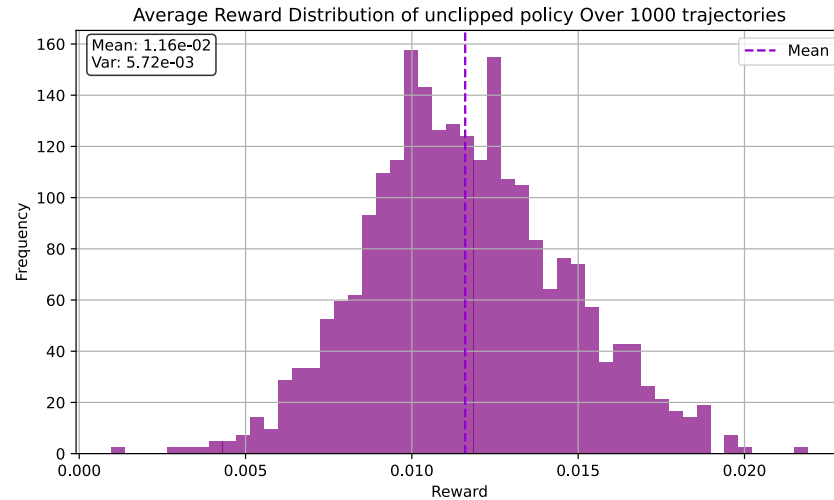


Figure 10: Unclipped baseline

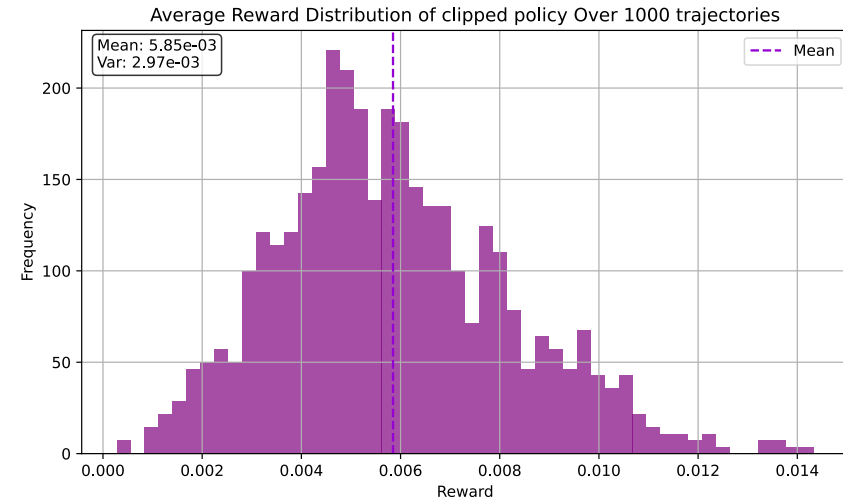


Figure 11: Clipped baseline